Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

TRAFFIC CONGESTION FORECASTING USING SENSOR DATA

The domain of the Project:
Data Science and Data Analytics

Team Mentors (and their designation):
Purnangshu Nath Roy (Trainer)

Team Members:
Ms. Shradha Pramod Pol

Period of the project

November 2025 to June 2026

Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Declaration

The project titled **"Traffic Congestion Forecasting using Sensor Data"** has been mentored by **Purnangshu Nath Roy**, organised by SURE Trust, from **November 2025 to June 2026**, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the member of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

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1. Executive Summary

Traffic congestion has become a major challenge in urban areas due to increasing vehicle density, changing weather conditions, and inefficient traffic management. The objective of this project was to develop an interactive Traffic Congestion Forecasting Dashboard using sensor-based traffic data to analyze traffic patterns, identify congestion hotspots, and provide meaningful insights for traffic monitoring and decision-making.

The project involved collecting and analyzing traffic-related data containing information such as traffic volume, vehicle type, road segments, weather conditions, congestion levels, speed, and sensor readings. The data was processed, cleaned, and transformed to create meaningful metrics and visualizations. Power BI was used as the primary business intelligence tool for dashboard development, enabling interactive exploration of traffic conditions through filters, charts, and analytical reports.

The dashboard was designed with multiple analytical pages, including Overview, Traffic Flow Analysis, Congestion Hotspots Analysis, and Weather Impact Analysis. These pages provide insights into peak traffic periods, weekday versus weekend traffic patterns, highly congested road segments, traffic severity by location, weather-related congestion impacts, and average speed variations under different conditions.

The analysis revealed that traffic congestion is significantly influenced by peak-hour traffic volume, weather conditions such as fog and rain, and specific road segments with high vehicle density. The dashboard enables stakeholders to quickly identify congestion trends, evaluate traffic performance, and support data-driven traffic management decisions.

The project successfully demonstrates the application of data analytics and visualization techniques in solving real-world transportation challenges. The developed dashboard provides an effective solution for monitoring traffic conditions and can serve as a foundation for future predictive traffic management systems and smart city initiatives.

2. Introduction

2.1 Background and Context of the Project:

Traffic congestion is one of the most significant challenges faced by modern cities. Rapid urbanization, increasing vehicle ownership, and limited road infrastructure have led to higher traffic density, resulting in delays, increased fuel consumption, environmental pollution, and reduced productivity. Effective traffic management requires continuous monitoring and analysis of traffic conditions to identify congestion patterns and support informed decision-making.

With the advancement of sensor technology and data analytics, large volumes of traffic-related data can now be collected and analyzed in real time. Business Intelligence tools such as Power BI provide powerful capabilities for transforming raw traffic data into meaningful visual insights, enabling traffic authorities and stakeholders to better understand traffic behavior and congestion trends.

2.2 Problem Statement:

Managing traffic congestion efficiently requires the ability to analyze multiple factors such as traffic volume, vehicle types, weather conditions, road segments, and congestion severity. Traditional monitoring methods often lack interactive visualization and comprehensive analytical capabilities, making it difficult to identify congestion hotspots and understand the factors influencing traffic flow.

This project addresses this challenge by developing a Traffic Congestion Forecasting Dashboard that provides a centralized platform for monitoring, analyzing, and visualizing traffic conditions using sensor-generated data.

2.3 Scope and Limitations:

2.3.1 Scope:

- Analyze traffic volume and congestion patterns using sensor data.
- Study the impact of weather conditions on traffic flow and congestion levels.
- Identify highly congested road segments and traffic hotspots.
- Compare traffic behavior across different locations and vehicle types.
- Develop an interactive dashboard for traffic monitoring and decision-making using Power BI.

2.3.2 Limitations:

- The project relies on historical sensor data and does not perform real-time traffic forecasting.
- The analysis is limited to the available dataset and selected traffic parameters.
- External factors such as accidents, road construction activities, and special events are not included in the analysis.

2.4 Innovation Component in the Project:

The innovative aspect of this project lies in the integration of traffic, weather, vehicle, and road segment data into a single interactive dashboard. Instead of presenting static reports, the solution enables dynamic exploration of traffic conditions through filters, visualizations, and analytical insights. The dashboard helps users quickly identify congestion trends, evaluate traffic performance, and understand the impact of various factors on traffic flow.

By combining data analytics with interactive visualization techniques, the project demonstrates how Business Intelligence solutions can support smarter traffic management and contribute to future smart city initiatives.

3. Project Objectives

The primary objective of this project is to develop an interactive Traffic Congestion Forecasting Dashboard using sensor-based traffic data to analyze traffic patterns, monitor congestion levels, and generate meaningful insights for effective traffic management.

3.1 Specific Objectives:

1. To analyze traffic volume patterns across different time periods, locations, and vehicle types.
2. To identify peak and off-peak traffic hours and evaluate their impact on traffic flow and average speed.
3. To study congestion levels and classify traffic conditions into low, medium, and high congestion categories.
4. To identify congestion hotspots by analyzing road segments, locations, and congestion severity scores.
5. To evaluate the influence of weather conditions such as fog, rain, cloudy, and clear weather on traffic congestion and vehicle speed.
6. To compare weekday and weekend traffic patterns for understanding variations in traffic behavior.
7. To develop interactive Power BI dashboards that provide dynamic filtering, visualization, and exploration of traffic data.
8. To transform raw traffic sensor data into actionable insights that support data-driven decision-making.
9. To demonstrate the application of data analytics and business intelligence techniques in solving real-world transportation and urban mobility challenges.
10. To provide a scalable analytical framework that can support future smart traffic monitoring and forecasting systems.

4. Methodology and Results

4.1 Methods / Technology Used:

The Traffic Congestion Forecasting Dashboard project was developed using a structured data analytics methodology that combined data preparation, database management, data processing, statistical analysis, and business intelligence reporting. The objective was to transform raw traffic sensor data into meaningful insights that could support traffic monitoring and congestion management.

The project followed a multi-stage workflow beginning with data preparation in Microsoft Excel, followed by data storage and querying in MySQL. The processed data was then analyzed using Python, where data cleaning, transformation, and analytical computations were performed. Various Data Science techniques were applied to study traffic patterns, congestion severity, weather impacts, and traffic flow behavior. Finally, Power BI was used to create interactive dashboards and visualizations that provide actionable insights into traffic congestion.

The methodology ensured that the data moved through a complete analytics pipeline, enabling accurate analysis and effective visualization of traffic conditions.

4.2 Tools / Software Used:

Several software tools and technologies were utilized during the project lifecycle.

- **Microsoft Excel**

Microsoft Excel was used during the initial stages of the project for data inspection, validation, cleaning, and exploratory analysis. Pivot tables, formulas, and charts were used to understand traffic trends and identify data quality issues.

- **MySQL**

MySQL served as the database management system for storing, organizing, and querying traffic data. Multiple tables including Traffic_Data, Sensors, and Vehicles were created and managed within the MySQL database. SQL queries were used for data aggregation, congestion analysis, and traffic classification. The MySQL database was successfully connected with Power BI to enable direct data retrieval and dashboard development.

- **Python**

Python was used for data preprocessing, normalization, feature engineering, and analytical computations. Python libraries enabled efficient handling of large datasets and helped automate the analysis process.

- **Pandas**

Pandas was used for loading, cleaning, transforming, and manipulating traffic datasets.

- **NumPy**

NumPy was utilized for numerical calculations and statistical operations.

- **Matplotlib**

Matplotlib was used to create visual representations of traffic trends and support exploratory data analysis.

- **Data Science Techniques**

Statistical analysis, trend analysis, congestion indexing, and weather impact analysis were performed to generate meaningful insights from the traffic dataset.

- **Power BI**

Power BI was used to develop interactive dashboards and reports. It enabled visualization of traffic flow patterns, congestion hotspots, weather impacts, and traffic severity metrics.

- **Power Query**

Power Query was used within Power BI for data transformation and loading.

- **DAX (Data Analysis Expressions)**

DAX measures and calculated columns were created to develop KPIs, traffic metrics, congestion indicators, and analytical calculations.

4.3 Data Collection Approach:

The project utilized a structured traffic sensor dataset containing information related to traffic flow and congestion conditions. The dataset included variables such as traffic volume, vehicle type, road segment, location, weather conditions, average speed, congestion level, severity score, and timestamp information.

The data collection process involved obtaining the traffic dataset and validating its structure using Microsoft Excel. The dataset was inspected for missing values, duplicate records, inconsistent formats, and outliers. After initial validation, the data was imported into MySQL for structured storage and management.

The organized dataset was exported and processed using Python, where additional transformations and feature engineering activities were performed. Statistical measures and traffic-related indicators were calculated to support subsequent analytical tasks.

The final processed dataset was then imported into Power BI for dashboard development and visualization.

4.4 Project Architecture:

4.4.1 Architecture Overview:

The project architecture follows a sequential analytics workflow that integrates multiple technologies to transform raw traffic sensor data into meaningful business insights.

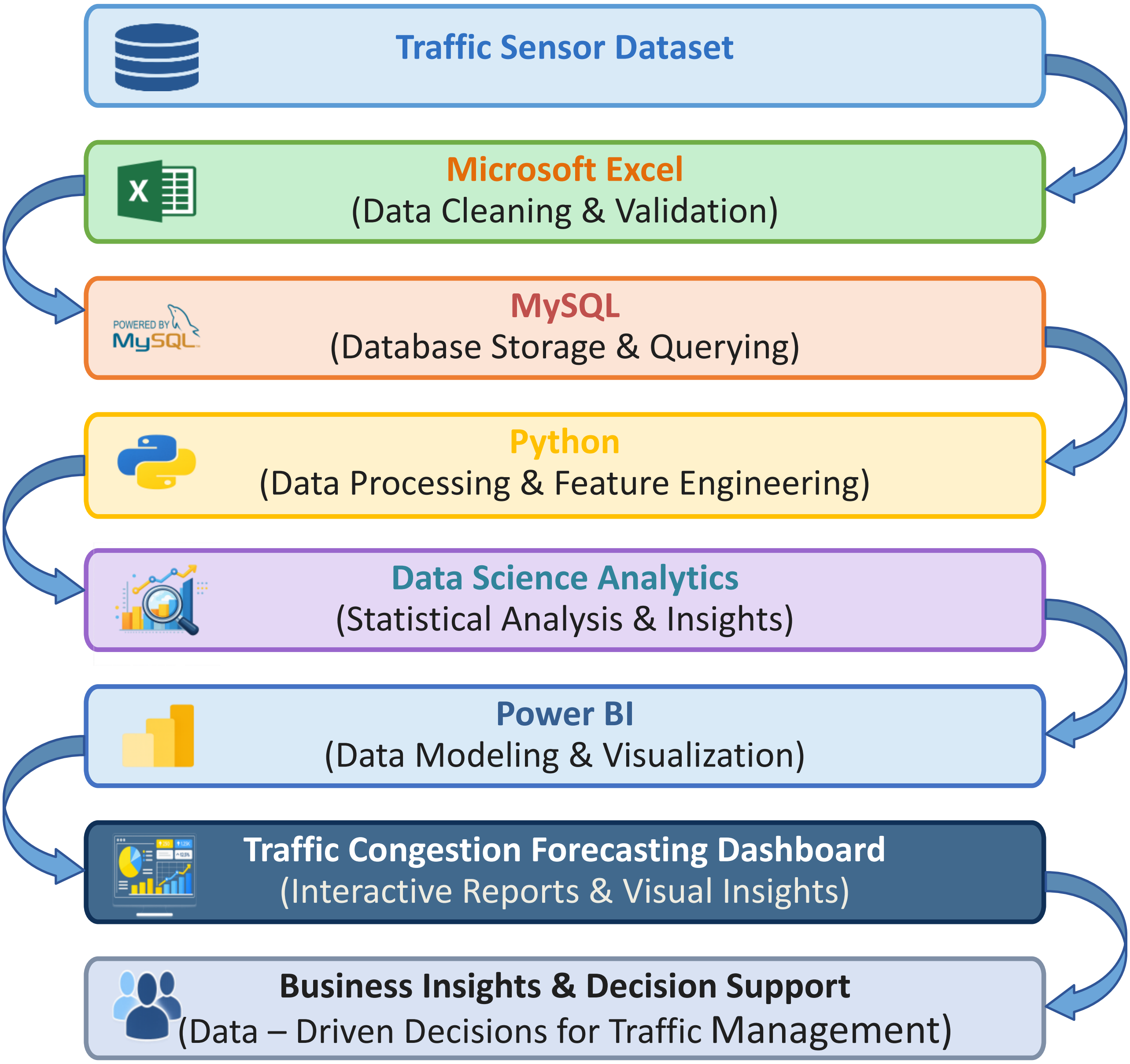

Figure 4.4.1: Project Architecture of Traffic Congestion Forecasting Dashboard

4.4.2 Architecture Description:

Figure 4.4.1. illustrates the architecture of the Traffic Congestion Forecasting Dashboard. The architecture follows a sequential analytics workflow that transforms raw traffic sensor data into meaningful business insights through multiple stages of data processing, analysis, and visualization.

The process begins with the Traffic Sensor Dataset, which serves as the primary source of data for the project. The dataset contains information related to traffic volume, average speed, vehicle type, road segment, location, weather conditions, congestion levels, and timestamps. These attributes provide the foundation for analyzing traffic behavior and congestion patterns.

The dataset is first processed using Advanced Excel, where preliminary data cleaning, validation, and exploratory analysis are performed. During this stage, missing values, duplicate records, formatting inconsistencies, and data quality issues are identified and corrected. Excel tools such as filters, formulas, and pivot tables are used to gain an initial understanding of traffic trends and data distribution.

After the data is validated, it is imported into MySQL for structured storage and management. MySQL enables efficient handling of large volumes of traffic data and supports the execution of SQL queries for data retrieval, filtering, aggregation, and summarization. This stage helps organize the dataset and prepare it for further analytical processing.

The processed data is then exported to Python, where advanced data preprocessing and feature engineering activities are performed. Python libraries such as Pandas and NumPy are used to clean the dataset, transform variables, create new analytical features, and prepare the data for statistical analysis. Additional traffic-related metrics are generated to enhance the quality of analysis.

The refined dataset is then subjected to Data Science Analytics. In this stage, statistical analysis techniques are applied to identify traffic patterns, evaluate congestion behavior, and analyze the impact of weather conditions on traffic flow. Congestion indices and severity scores are calculated to quantify traffic conditions and support deeper analytical insights.

The analytical dataset is subsequently imported into Power BI through a database connection with MySQL, where data modeling and visualization activities are carried out. Power Query is used for data transformation and data preparation, while DAX measures are created to calculate key performance indicators (KPIs) and business metrics. Interactive visualizations are developed to present traffic information in an intuitive and user-friendly manner, enabling users to monitor traffic patterns, analyze congestion levels, and derive meaningful insights from the processed dataset.

The Power BI solution includes multiple analytical dashboards such as Executive Overview, Traffic Flow Analysis, Congestion Hotspots Analysis, Weather Impact Analysis, and a Road Segment Drill-Through Report. The drill-through functionality enables users to navigate from summary-level reports to detailed road-segment analysis, supporting deeper exploration of traffic conditions.

To enhance dashboard security and data governance, Row-Level Security (RLS) was implemented using Admin, Traffic_Manager, and Viewer roles. This role-based access control mechanism demonstrates how different users can access dashboard information according to predefined permissions.

The output of the Power BI development stage is the Traffic Congestion Forecasting Dashboard, which provides a comprehensive platform for monitoring and analyzing traffic conditions. The dashboard consists of multiple analytical pages, including Executive Overview, Traffic Flow Analysis, Congestion Hotspots Analysis, and Weather Impact Analysis. These dashboards enable users to explore traffic patterns through interactive charts, KPIs, filters, and visual reports.

Finally, the dashboard generates Business Insights and Decision Support, which represent the ultimate objective of the project. The insights obtained from the dashboard help identify congestion hotspots, understand traffic flow behavior, evaluate weather-related impacts, monitor traffic severity levels, and support data-driven traffic management decisions. These insights can assist transportation authorities and planners in improving road network efficiency and reducing traffic congestion.

4.5 Final Project Working Screenshots Along with Supporting Explanation:

4.5.1 Dashboard Page 1 – Executive Overview

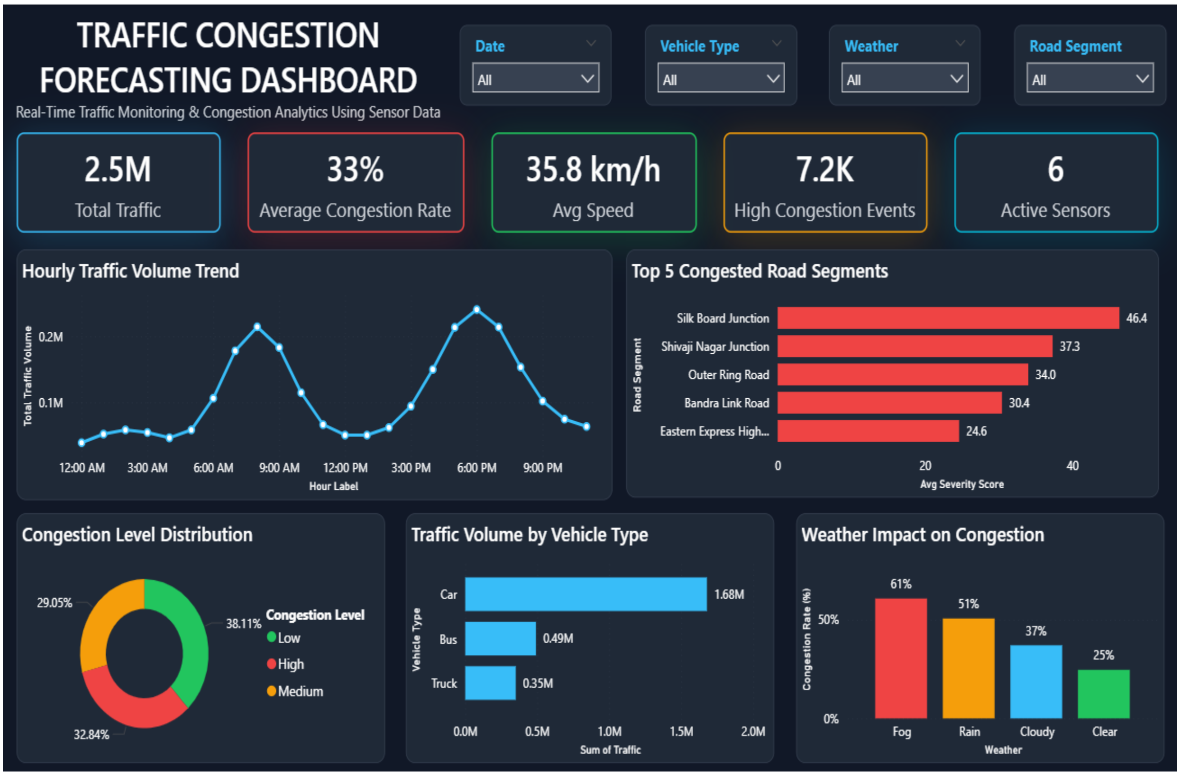

Figure 4.5.1: Executive Overview Dashboard

The Executive Overview dashboard provides a summary of overall traffic conditions and congestion performance. It displays key performance indicators (KPIs) such as Total Traffic Volume, Average Congestion Rate, Average Speed, High Congestion Events, and Active Sensors, enabling users to quickly assess traffic performance.

The dashboard includes an Hourly Traffic Volume Trend chart to identify peak traffic periods and a Top Congested Road Segments chart to highlight critical congestion areas. Additional visualizations present congestion level distribution, traffic volume by vehicle type, and the impact of weather conditions on congestion, supporting effective traffic monitoring and decision-making.

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4.5.2 Dashboard Page 2 – Traffic Flow Analysis

Figure 4.5.2: Traffic Flow Analysis Dashboard

The Traffic Flow Analysis dashboard focuses on understanding traffic movement patterns across different time periods and locations. It compares traffic volume during peak and off-peak hours, helping identify periods of high traffic demand. The weekday and weekend traffic comparison provides insights into variations in travel behavior throughout the week.

The dashboard also presents daily traffic volume trends, average vehicle speeds during peak and off-peak periods, and the top traffic-generating locations. Additionally, the Traffic Cluster Distribution categorizes traffic conditions into Low, Medium, and High traffic levels, enabling better understanding of traffic intensity patterns. These insights support traffic planning, congestion management, and operational decision-making.

4.5.3 Dashboard Page 3 – Congestion Hotspots Analysis

Figure 4.5.3: Congestion Hotspots Analysis Dashboard

The Congestion Hotspots Analysis dashboard is designed to identify locations and road segments experiencing significant traffic congestion. The Congestion Hotspot Matrix uses conditional formatting to visually highlight congestion severity across different cities and road segments, allowing users to quickly identify critical congestion hotspots.

Additional visualizations analyze congestion severity by location and compare congestion levels across different road segment types such as intersections, urban arterials, and highways. The dashboard also includes a severity distribution chart that categorizes congestion events into Low, Medium, and High severity levels. These insights help traffic authorities prioritize congestion mitigation strategies and improve traffic management efficiency.

4.5.4 Dashboard Page 4 – Weather Impact Analysis

Figure 4.5.4: Weather Impact Analysis Dashboard

The Weather Impact Analysis dashboard examines how different weather conditions influence traffic flow, congestion levels, and vehicle speed. The dashboard compares congestion rates under Fog, Rain, Cloudy, and Clear weather conditions, helping identify weather scenarios that contribute to increased traffic congestion.

It also analyzes traffic volume by weather condition, average speed variations, and weather severity levels. The weather condition distribution chart provides an overview of the frequency of different weather patterns within the dataset. Together, these visualizations help understand the relationship between weather and traffic behavior, supporting better traffic forecasting and operational planning.

4.5.5 Dashboard Page 5 – Road Segment Drill-Through Report

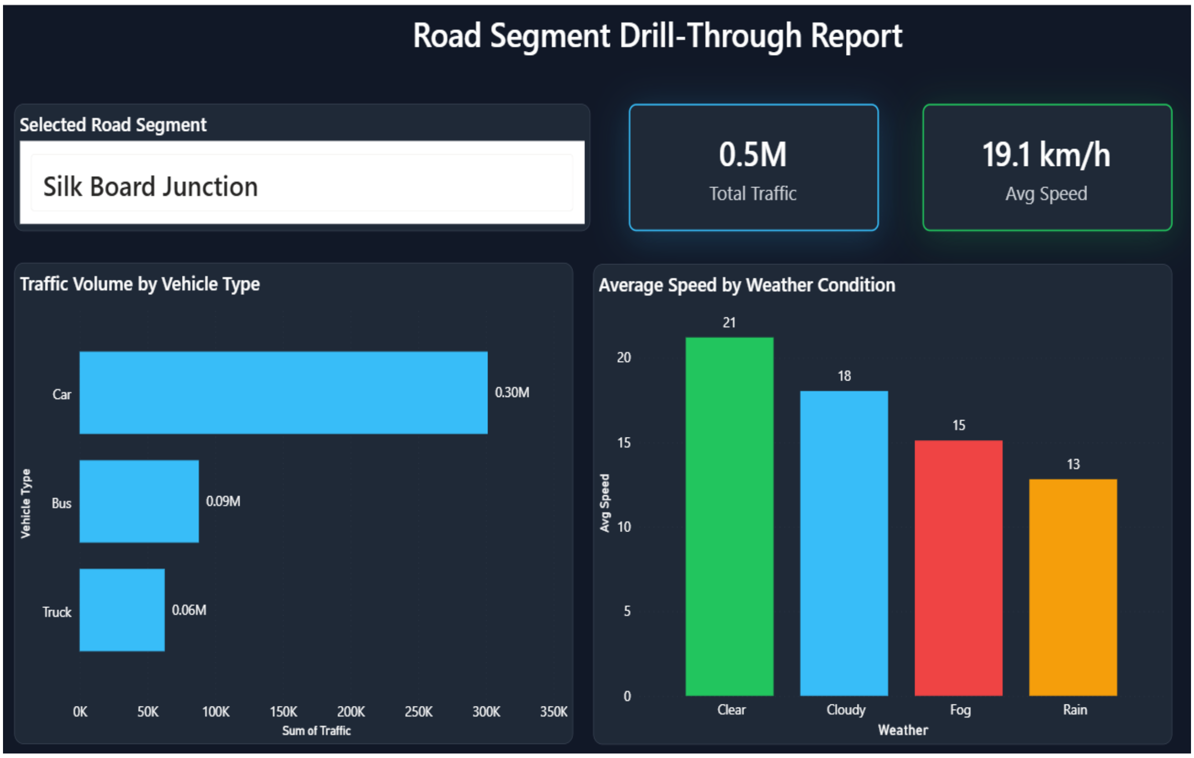

Figure 4.5.5: Road Segment Drill-Through Report

The Road Segment Drill-Through Report provides detailed insights into individual road segments using Power BI's drill-through functionality. Users can navigate from the Overview Dashboard to analyze a specific road segment in greater detail.

The report displays key performance indicators such as Total Traffic and Average Speed for the selected road segment. It also includes visualizations showing Traffic Volume by Vehicle Type and Average Speed by Weather Condition, helping users understand traffic patterns and the impact of weather on road performance.

This report supports road-level traffic analysis and enables better monitoring of congestion and traffic conditions.

4.6 Security Implementation (Row-Level Security):

Row-Level Security (RLS) was implemented in Power BI to demonstrate role-based access control and secure dashboard access. Three user roles were created: Admin, Traffic_Manager, and Viewer. The roles were configured using the Manage Roles feature and validated using the View As Roles option. This implementation helps ensure that users can access dashboard data according to their assigned permissions and demonstrates basic security management within Power BI.

4.6.1 Row-Level Security Roles Configuration

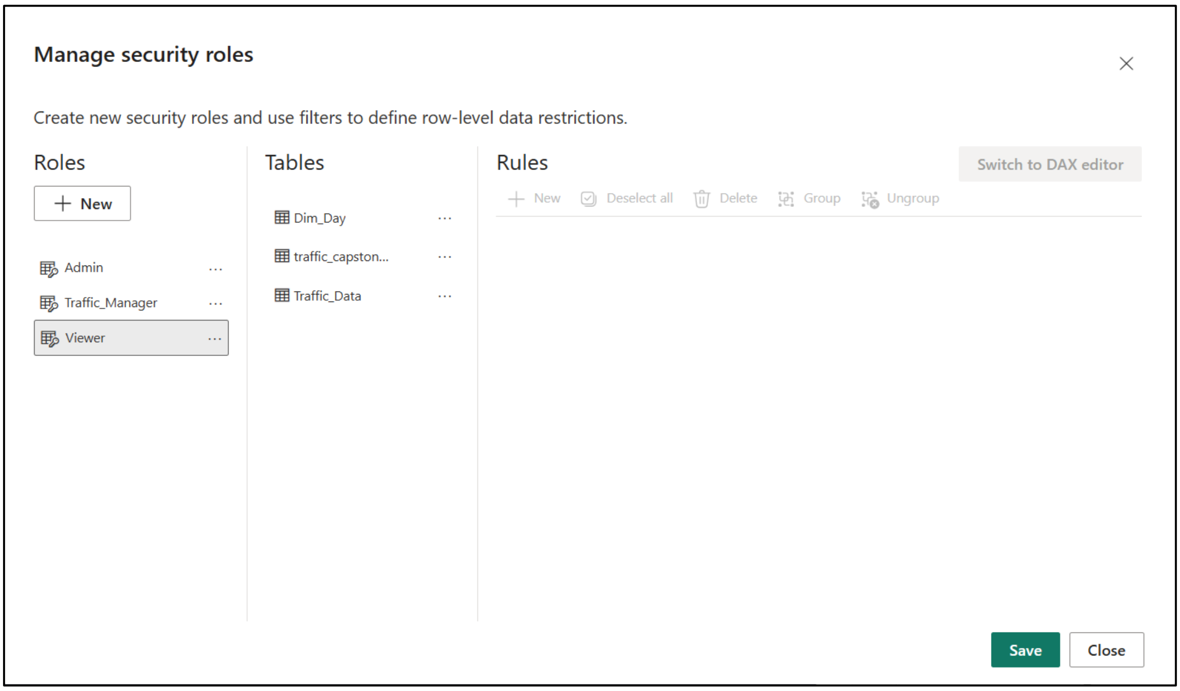

Figure 4.6.1: Row-Level Security Roles Configuration

The Manage Roles feature in Power BI was used to create and configure multiple security roles, including Admin, Traffic_Manager, and Viewer. These roles represent different levels of user access and help implement role-based security within the dashboard. By defining separate roles, the system ensures that users can access and interact with data according to their responsibilities and permissions.

4.6.2 Testing Row-Level Security Using View As Roles

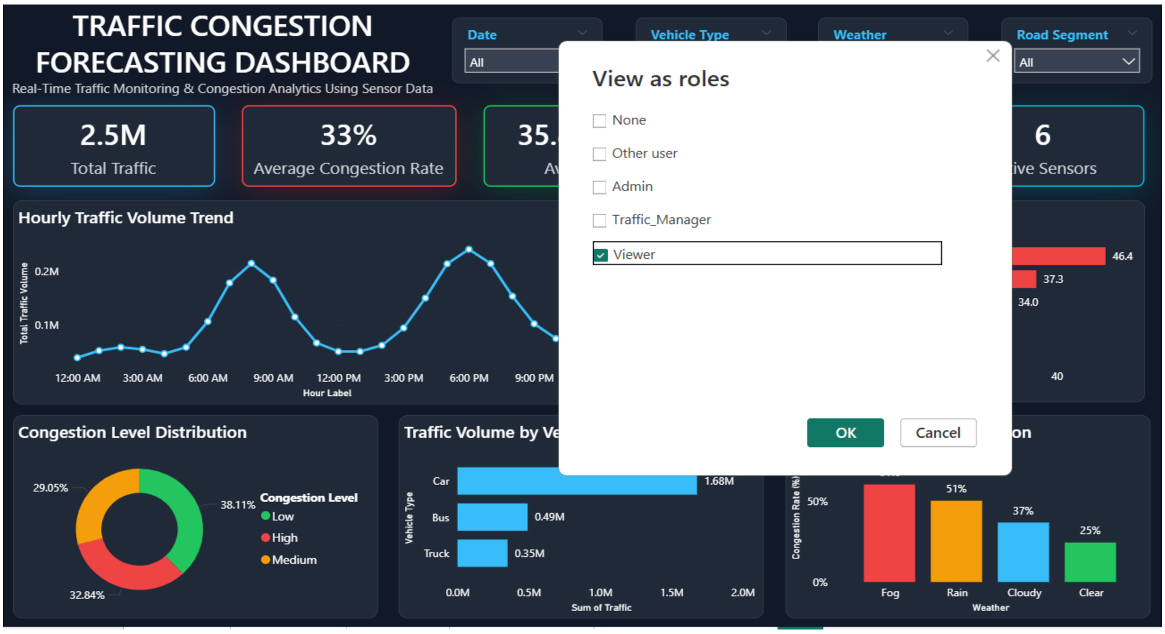

Figure 4.6.2: Testing Row-Level Security Using View As Roles

The View As Roles feature was used to test and verify the configured security roles. This functionality allows developers to simulate different user roles and validate that data access restrictions are working correctly. The testing process confirms that users can only view the data permitted by their assigned role, thereby improving dashboard security and data confidentiality.

5. Social and Industry Relevance of the Project

5.1 Social Relevance:

Traffic congestion is a growing concern in urban areas due to increasing vehicle populations and limited transportation infrastructure. Congestion leads to longer travel times, increased fuel consumption, environmental pollution, and reduced productivity. The Traffic Congestion Forecasting Dashboard helps address these challenges by providing a data-driven platform for monitoring and analyzing traffic conditions.

The dashboard enables users to identify congestion hotspots, understand traffic flow patterns, and evaluate the impact of weather conditions on road traffic. By providing clear and interactive visual insights, the system supports better traffic planning and management, which can contribute to reduced congestion, improved commuter experiences, and enhanced road safety.

The project also promotes the use of data analytics and business intelligence technologies in solving real-world transportation problems. It demonstrates how analytical tools can transform raw traffic data into meaningful information that benefits society and supports sustainable urban development.

5.2 Industry Relevance:

The project has significant relevance in the transportation, logistics, smart city, and urban planning sectors. Government agencies, municipal corporations, traffic management authorities, and transportation planners can utilize similar analytical solutions to monitor traffic conditions and make informed decisions.

The dashboard provides valuable insights into traffic volume, congestion severity, weather impacts, and location-based traffic behavior. Such information can help organizations optimize traffic operations, improve road network efficiency, allocate resources effectively, and develop congestion mitigation strategies.

In the context of smart city initiatives, the project demonstrates how data analytics, database technologies, Python-based analysis, and Power BI dashboards can be integrated to support intelligent transportation systems. The analytical framework developed in this project can serve as a foundation for future real-time traffic monitoring, predictive traffic forecasting, and advanced mobility management solutions.

6. Learning and Reflection

6.1 New Learnings During the Project:

This project provided practical exposure to the complete data analytics lifecycle, starting from data collection and preprocessing to dashboard development and business insight generation. Through this project, I enhanced my knowledge of Advanced Excel for data cleaning, validation, and exploratory analysis. I gained hands-on experience in MySQL by writing queries for data extraction, filtering, aggregation, and analysis.

The project also strengthened my Python programming skills, particularly in data preprocessing, feature engineering, statistical analysis, and congestion-related calculations using libraries such as Pandas and NumPy. Furthermore, I learned how to apply data science techniques to identify traffic patterns, analyze weather impacts, and generate meaningful insights from large datasets.

In Power BI, I developed expertise in data modeling, Power Query transformations, DAX calculations, KPI development, interactive dashboard design, conditional formatting, and report navigation. Overall, the project improved my analytical thinking, problem-solving abilities, and data visualization skills.

6.2 Overall Experience:

Working on the Traffic Congestion Forecasting Dashboard was a valuable learning experience. The project helped me understand how different technologies such as Excel, MySQL, Python, Data Science, and Power BI can be integrated into a complete analytics solution. I faced several challenges related to data cleaning, dashboard design, DAX calculations, and visualization formatting, which improved my troubleshooting and problem-solving capabilities.

The project also enhanced my ability to convert raw data into actionable business insights and present them through professional dashboards. Overall, the experience significantly improved my technical knowledge, analytical skills, and confidence in handling real-world data analytics projects.

7. Conclusion and Future Scope

7.1 Conclusion:

The Traffic Congestion Forecasting Dashboard was successfully developed to analyze traffic flow patterns, congestion hotspots, vehicle movement, and weather impacts using sensor-based traffic data. The project integrated multiple technologies including Excel, MySQL, Python, Data Science techniques, and Power BI to transform raw traffic data into meaningful insights.

The dashboard provides an effective platform for monitoring traffic conditions through interactive visualizations, KPIs, and analytical reports. Key findings revealed the influence of peak traffic periods, location-based congestion patterns, road segment severity, and weather conditions on overall traffic performance. The developed solution supports data-driven decision-making and demonstrates the practical application of business intelligence and analytics in traffic management.

7.2 Future Scope:

The project can be further enhanced by integrating real-time traffic sensor data and live weather APIs for continuous monitoring and forecasting. Machine Learning models can be incorporated to predict future congestion levels and traffic volumes with higher accuracy. Additional features such as route optimization, accident prediction, anomaly detection, and mobile dashboard accessibility can further improve the usefulness of the system.

The dashboard can also be expanded to support smart city initiatives by integrating data from GPS devices, IoT sensors, and public transportation systems. These enhancements would enable more advanced traffic management and decision-support capabilities.